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ROTARY JOINT DEVICE, MOTOR, AND MACHINE TOOL

Abstract

The rotary joint device includes: a rotary-side joint having a first hole through which liquid is to flow in an axial direction; a fixed-side joint configured to reciprocate in the axial direction of approaching and separating from the rotary-side joint, the fixed-side joint further having a second hole in the axial direction that communicates with the first hole in a state where the fixed-side joint is in contact with the rotary-side joint; an attachment being fixable to another member; a DLC (diamond-like carbon) film provided on an outer peripheral surface of the fixed-side joint; and an elastic member interposed liquid-tight between an inner peripheral surface of the attachment and the DLC film.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-037360 filed on Mar. 11, 2024. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

[0002] Conventionally, for example, a machine tool may be provided with a rotary joint device. When the rotary joint device includes respective tubular rotary-side joint and fixed-side joint, the end face of one end of the rotary-side joint faces the end face of one end of the fixed-side joint. The other end of the rotary-side joint is connected to a tubular rotary shaft provided by, for instance, a motor. The attachment provided by the rotary joint device is fixed to a fixed member provided by the machine tool. The attachment has a through hole. The other end of the fixed-side joint is inserted into the through hole of the attachment. Coolant (liquid) is pumped by the pump of a coolant supply device installed outside the machine tool and flows through the through hole of the attachment. An elastic member is interposed liquid-tight (to prevent liquid leakage) between the inner peripheral surface of the through hole of the attachment and the outer peripheral surface of the fixed-side joint.

[0003] The rotary shaft of the motor rotates in the circumferential direction. The rotation of the rotary shaft is transmitted to the tool, and the tool processes the workpiece. The tool has a nozzle.

[0004] By the coolant flowing through the through hole of the attachment pushing the fixed-side joint, the fixed-side joint comes into contact with the rotary-side joint. The coolant flows through the through hole of the attachment, the through hole of the fixed-side joint, the through hole of the rotary-side joint, and the through hole of the rotary shaft in this order, and is ejected towards the workpiece from the nozzle provided by the tool.

[0005] As the rotary shaft rotates, the rotary-side joint rotates in the circumferential direction. If the supply of coolant to the through hole of the attachment stops while the motor is rotating, the fixed-side joint will be separated from the rotary-side joint due to the wind pressure created by the rotation of the rotary-side joint or because the biasing member of the rotary joint device pushes the fixed-side joint.

[0006] In a conventional rotary joint device, the fixed-side joint moves back and forth in the direction of approaching and separating from the rotary-side joint. Due to the reciprocating movement of the fixed-side joint, the elastic member slides relative to the outer peripheral surface of the fixed-side joint.

[0007] In such conventional rotary joint device, supply holes may be provided for supplying grease between the outer peripheral surface of the fixed-side joint (referred to as “fixed shaft” in the document) and the elastic member (referred to as “O-ring” in the document). With the intervention of grease, the reciprocating movement of the fixed-side joint is smooth.

[0008] However, if the contact of the fixed-side joint with the rotary-side joint is not smooth, the coolant might leak between the fixed-side joint and the rotary-side joint. If the separation of the fixed-side joint from the rotary-side joint is not smooth, the fixed-side joint and the rotary-side joint might wear unnecessarily.

SUMMARY

[0009] Thus configured conventional rotary joint device is rather complicated to the extent that it requires grease supply holes.

[0010] An object of the present disclosure is to provide a rotary joint device, a motor, and a machine tool that can achieve both smooth reciprocating movement of the fixed-side joint and simplification of the structure.

[0011] The rotary joint device according to the present disclosure includes: a tubular rotary-side

joint having a first hole through which liquid is to flow in an axial direction, the rotary-side joint being rotatable in a circumferential direction; a fixed-side joint having a tubular shape and an end face at one end facing an end face at one end of the rotary-side joint, the fixed-side joint being configured to reciprocate in the axial direction of approaching and separating from the rotary-side joint, the fixed-side joint further having a second hole in the axial direction that communicates with the first hole of the rotary-side joint in a state where the fixed-side joint is in contact with the rotary-side joint; an attachment having an inner peripheral surface, the attachment being fixable to another member, the inner peripheral surface defining a third hole through which the liquid is to flow and into which the other end of the fixed-side joint is inserted; a DLC (diamond-like carbon) film provided on an outer peripheral surface of the fixed-side joint; and an elastic member interposed liquid-tight between the inner peripheral surface of the attachment and the DLC film, the elastic member being slidable with respect to the DLC film when the fixed-side joint moves in the axial direction with respect to the rotary-side joint. The DLC film is provided in a contact area where the elastic member contacts on the outer peripheral surface.

[0012] In the rotary joint device according to the present disclosure, both the rotary-side joint and the fixed-side joint may be configured to have tubular shape.

[0013] The rotary-side joint may be rotatable in the circumferential direction. The fixed-side joint may be configured to be capable of reciprocating in the axial direction of approaching and separating from the rotary-side joint. The end face of one end of the fixed-side joint may face the end face of one end of the rotary-side joint. The fixed-side joint may have the second hole in the axial direction.

[0014] The attachment may be used by being fixed to another member. The attachment may have the third hole. The liquid may flow through the third hole. Since the other end of the fixed-side joint is inserted into the third hole, the third hole and the second hole of the fixed-side joint may communicate with each other.

[0015] A DLC film may be provided on the outer peripheral surface of the fixed-side joint. An elastic member may be interposed liquid-tight between the inner peripheral surface of the third hole of the attachment and the DLC film. The elastic member may slide relatively to the DLC film when the fixed-side joint moves in the axial direction. Hereinafter, the relative sliding of the elastic member against the DLC film is simply referred to as the sliding of the elastic member.

[0016] The DLC film may be provided in the contact area. The contact area may be in the range on the outer peripheral surface of the fixed-side joint where the elastic member contacts. Due to the low friction of the DLC film against the elastic member, the sliding of the elastic member can be smooth. Therefore, the movement of the fixed-side joint in the direction of approaching and separating from the rotary-side joint can be smooth.

[0017] Since the DLC film may be provided on the outer peripheral surface of the fixed-side joint using a well-known DLC coating technique, it is not necessary to form supply holes in the attachment for supplying grease. Therefore, the structure of the rotary joint device is simplified.

[0018] As a result, smooth reciprocating movement of the fixed-side joint and simplification of the structure can be achieved simultaneously.

[0019] The rotary-side joint may have the first hole in the axial direction through which fluid flows. When the fixed-side joint is in contact with the rotary-side joint, the second hole of the fixed-side joint may communicate with the first hole of the rotary-side joint. The liquid may flow through the third hole of the attachment, the second hole of the fixed-side joint, and the first hole of the rotary-side joint in this sequence.

[0020] Since the frictional force generated between the DLC film and the elastic member is small, the rotary joint device can be designed so that the force with which the elastic member contacts the DLC film is large. Therefore, liquid leakage between the inner peripheral surface of the third hole of the attachment and the outer peripheral surface of the fixed-side joint can be reliably prevented.

[0021] In the rotary joint device according to the present disclosure, the DLC film may be provided

in the contact area and a non-contact area on the outer peripheral surface where the elastic member does not contact, and the non-contact area may be located on the end face side of the other end of the fixed-side joint relative to the contact area.

[0022] In the rotary joint device according to the present disclosure, the DLC film may be provided in the contact area and the non-contact area. The non-contact area may be a region on the outer peripheral surface of the fixed-side joint where the elastic member does not contact and is located on the end face side of the other end relative to the contact area.

[0023] A part of the liquid flowing through the third hole of the attachment may infiltrate between the DLC film provided in the non-contact area and the inner peripheral surface of the third hole. If foreign matter is mixed in the liquid, the foreign matter may infiltrate between the inner peripheral surface of the third hole of the attachment and the DLC film. However, due to the low friction of the DLC film against the foreign matter, the foreign matter will not hinder the movement of the fixed-side joint in the axial direction.

[0024] The DLC film may not necessarily be provided in the range on the outer peripheral surface of the fixed-side joint from the lower end of the contact area to the lower end of the fixed-side joint. This is because liquid will not infiltrate (and hence foreign matter mixed in the liquid will not infiltrate) between the inner peripheral surface of the third hole provided in the attachment and the DLC film if the DLC film is provided in this range.

[0025] In the rotary joint device according to the present disclosure, the elastic member may be an O-ring made of fluororubber.

[0026] Since the friction coefficient between the O-ring made of fluororubber and the DLC film is sufficiently small, the sliding of the elastic member may become even smoother.

[0027] A motor according to the present disclosure includes a rotary shaft rotatable in the circumferential direction and having a tubular shape and a through hole through which liquid is to flow in the axial direction, wherein the rotary shaft is connected with the rotary-side joint of the rotary joint device according to the present disclosure at another end of the rotary-side joint which is opposite to the one end of the rotary-side joint to integrally rotate with the rotary-side joint while the through hole and the first hole of the rotary-side joint communicate with each other.

[0028] In the motor according to the present disclosure, the rotary-side joint of the rotary joint device is connected to the rotary shaft. Therefore, smooth reciprocating movement of the fixed-side joint and simplification of the structure can be achieved simultaneously.

[0029] The rotary shaft may have tubular shape and configured to be rotatable in the circumferential direction. The rotary shaft may have a through hole in the axial direction through which fluid flows. The rotary shaft may be connected at the other end of the rotary-side joint such that the through hole of the rotary shaft and the first hole of the rotary-side joint may communicate with each other, and the rotary shaft may rotate integrally with the rotary-side joint.

[0030] A machine tool according to the present disclosure is a machine tool for processing a workpiece while discharging coolant from a nozzle using a tool provided with the nozzle, the machine tool including: the rotary joint device according to the disclosure; a fixed member to which the attachment of the rotary joint device is fixed; and a motor including a rotary shaft rotatable in the circumferential direction and having a tubular shape and a fourth hole through which liquid is to flow in an axial direction, wherein the rotary shaft is connected with the rotary-side joint of the rotary joint device at another end of the rotary-side joint which is opposite to the one end of the rotary-side joint to integrally rotate with the rotary-side joint while the through hole and the first hole of the rotary-side joint communicate with each other, wherein rotation of the rotary shaft is to be transmitted to the tool to rotate the tool, wherein the liquid is to flow through the third hole, the second hole, the first hole, and the fourth hole in this sequence to be ejected through the nozzle of the tool, wherein the DLC film being provided in the contact area where the elastic member contacts on the outer peripheral surface.

[0031] The attachment of the rotary joint device according to the present disclosure is fixed to the

fixed member provided by the machine tool, and the rotary-side joint of the rotary joint device according to the present disclosure is connected to the rotary shaft of the motor provided by the machine tool. Therefore, smooth reciprocating movement of the fixed-side joint and simplification of the structure can be achieved simultaneously.

[0032] The rotary shaft may be configured to have a tubular shape and to be rotatable in the circumferential direction. The rotary shaft may have a fourth hole in the axial direction through which fluid flows. The rotary shaft may be connected to the other end of the rotary-side joint such that the fourth hole of the rotary shaft and the first hole of the rotary-side joint may communicate with each other, and the rotary shaft rotates integrally with the rotary-side joint. The rotation of the rotary shaft is transmitted to the tool, resulting in the rotation of the tool. The tool processes the workpiece. The tool has a nozzle. When the fixed-side joint is in contact with the rotary-side joint, the second hole of the fixed-side joint communicates with the first hole of the rotary-side joint. The liquid flows through the attachment's third hole, the fixed-side joint's second hole, the rotary-side joint's first hole, and the rotary shaft's fourth hole in this order and is ejected outward through the nozzle of the tool. The machine tool can process the workpiece while discharging coolant from the nozzle of the tool.

[0033] According to the rotary joint device, motor, and machine tool of the present disclosure, smooth reciprocating movement of the fixed-side joint and simplification of the structure can be achieved simultaneously.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0034] FIG. 1 is a perspective view of a machine tool according to an embodiment.

[0035] FIG. 2 is a schematic cross-sectional view of a main part of the spindle head.

[0036] FIG. 3 is an enlarged schematic cross-sectional view of a main part of the spindle head.

[0037] FIG. 4 is a cross-sectional view of the rotary joint device.

DESCRIPTION

[0038] Hereinafter, an embodiment of the present disclosure will be described. In the following description, the vertical (up-down), front-rear, and left-right directions indicated by the arrows in the drawings are used. As described later, in this embodiment, the upper side is the coolant (liquid) inflow side, and the lower side is the coolant outflow side.

[0039] FIG. 1 is a perspective view of a machine tool according to an embodiment. In the following, the left-right direction, the front-rear direction, and the vertical direction of the machine tool 1 are the X-axis direction, the Y-axis direction, and the Z-axis direction, respectively. The machine tool 1 includes a base 11, a column 12, and a spindle head 13. The base 11 is a substantially rectangular parallelepiped iron pedestal installed on the floor. The column 12 is provided at the upper rear part of the base 11. The spindle head 13 is provided on the front face of the column 12 so as to be movable in the vertical direction. The column 12 includes a Z-axis motor for moving the spindle head 13 in the Z direction.

[0040] FIG. 2 is a schematic cross-sectional view of a main part of the spindle head 13. The spindle head 13 includes a motor 21. The motor 21 includes a rotary shaft 22, a rotor and a stator.

[0041] The rotary shaft 22 is cylindrical. The rotary shaft 22 is rotatable in the circumferential direction. The axial direction of the rotary shaft 22 is vertical. Since the rotary shaft 22 is tubular, it has a coaxial through hole in the axial direction. This through hole is the fourth hole 220 (flow passage). By supplying power to the motor 21 from the power supply in the control box 17, the rotary shaft 22 rotates in the circumferential direction.

[0042] The spindle head 13 further includes a fixed member 23 and a spindle 24. The fixed member 23, for example, is a bottomed cylindrical shape. The bottomed cylindrical fixed member

23 is arranged coaxially with the rotary shaft **22**. The fixed member **23** is located above the motor **21**, with the opening of the fixed member **23** facing downward. The fixed member **23** includes a supply port **231**. The supply port **231**, for example, is located on the peripheral wall of the fixed member **23**.

[0043] The spindle **24** is located below the motor **21** and is at the lower part of the spindle head **13**. The axial direction of the spindle **24** is vertical. The spindle **24** has a coaxial through hole **240**. The upper end of the spindle **24** is connected coaxially and liquid-tight to the lower end of the rotary shaft **22** of the motor **21**. For example, a coupling **25** connects the rotary shaft **22** and the spindle **24** to each other. By connecting the rotary shaft **22** and the spindle **24**, the fourth hole **220** of the rotary shaft **22** and the through hole **240** of the spindle **24** communicate with each other, and the spindle **24** rotates in the circumferential direction along with the rotation of the rotary shaft **22**.

[0044] In the drawing, **20** is a tool, which, for example, is an end mill used to cut the workpiece **W**. The tool **20** is columnar. The tool **20** has a flow path **201**. The flow path **201** extends from one end of the tool **20** to the other end. The axial direction of the tool **20** is vertical. The lower end of the spindle **24** holds the upper end of the tool **20** detachably and liquid-tight. By the spindle **24** holding the tool **20**, the through hole **240** of the spindle **24** communicates with the flow path **201** of the tool **20**, and the tool **20** rotates in the circumferential direction along with the rotation of the spindle **24**. The tool **20** has at least one nozzle **202**. The nozzle **202** is located at the lower end of the tool **20** and communicates with the flow path **201**.

[0045] As shown in FIG. **1**, the machine tool **1** includes a tool changer **14**, frames **15**, a table **16**, and a control box **17**.

[0046] The tool changer **14** includes a disc-shaped magazine **141**. A pair of left and right frames **15** holds the magazine **141** on the front side of the column **12** so that it does not interfere with the spindle head **13**. The magazine **141** has multiple grip arms **142** arranged radially on the periphery. The grip arms **142** detachably hold the tool **20** through a tool holder. The tool changer **14** rotates the magazine **141** to position a predetermined tool **20** at the tool change position. The tool changer **14** replaces the tool **20** mounted on the spindle **24** with the tool **20** at the tool change position. The tool change position is the lowest position of the magazine **141**.

[0047] The table **16** is located in front of the column **12** and below the magazine **141** at the upper part of the base **11**. The user fixes the workpiece **W** to the table **16**. The base **11** supports the table **16** so that it can move in the X-axis direction and the Y-axis direction, respectively. The table **16** is moved in the X-axis direction and Y-axis direction by an X-axis motor and a Y-axis motor which are provided in the machine tool **1**.

[0048] The base **11** supports the control box **17** at the rear side of the column **12**. A controller provided in the control box **17** controls the operation of the machine tool **1**.

[0049] The table **16** moves forward, backward, left, and right to adjust the front-rear and left-right position of the tool **20** held by the spindle **24**. The spindle head **13** moves up and down to adjust the vertical position of the tool **20** held by the spindle **24**.

[0050] The machine tool **1** processes the workpiece **W** using the rotating tool **20**. The tool **20** cuts the workpiece **W** by contacting it while rotating. During the processing of the workpiece **W**, coolant is supplied from a coolant supply device to the internal space of the fixed member **23** through the supply port **231**, which increases the lubrication between the tool **20** and the workpiece **W**, and efficiently discharges chips.

[0051] As shown in FIG. **2**, the machine tool **1** further includes a rotary joint device **3**. FIG. **3** is an enlarged schematic cross-sectional view of a main part of the spindle head **13**, mainly showing the rotary joint device **3**. FIG. **4** is a cross-sectional view of the rotary joint device **3**. As shown in FIGS. **2-4**, the rotary joint device **3** includes an attachment **31**, a fixed-side joint **32**, and a rotary-side joint **33**.

[0052] The attachment **31** is tubular. Since the attachment **31** is tubular, it has a coaxial through hole in the axial direction. This through hole is the third hole **310**. The attachment **31** has a flange

311 on the outer peripheral surface of one end. The attachment **31** is fixed to the fixed member **23**. [0053] For example, the attachment **31** is oriented vertically with a flange **311** at the lower end. The attachment **31** is inserted liquid-tight from below into the fixed member **23** through an opening in the fixed member **23**. The flange **311** of the attachment **31** contacts the lower end face of the fixed member **23**. By being inserted into the fixed member **23**, the third hole **310** of the attachment **31** communicates with the internal space of the fixed member **23** through the supply port **231**. The attachment **31** is located below the supply port **231** of the fixed member **23** and does not block the supply port **231**.

[0054] As shown in FIGS. **3** and **4**, the attachment **31** has a groove **312** on its outer peripheral surface. The groove **312** runs continuously in the circumferential direction of the attachment **31**. An O-ring **261** is housed in the groove **312**. As shown in FIG. **3**, the O-ring **261** is interposed over the entire circumference between the outer peripheral surface of the attachment **31** and the inner peripheral surface of the fixed member **23**. Therefore, liquid leakage between the attachment **31** and the fixed member **23** can be prevented.

[0055] As shown in FIGS. **2-4**, both the fixed-side joint **32** and the rotary-side joint **33** are tubular. The inner diameter and outer diameter of the fixed-side joint **32** and the inner diameter and outer diameter of the rotary-side joint **33** are approximately the same. Both the fixed-side joint **32** and the rotary-side joint **33** are made of, for example, iron-based metals such as stainless steel or chrome-molybdenum steel, or aluminum-based metals. The axial direction of both the fixed-side joint **32** and the rotary-side joint **33** is vertical. The fixed-side joint **32** and the rotary-side joint **33** are arranged coaxially in this order from the top. The fixed-side joint **32** can reciprocate vertically relative to the attachment **31** but does not rotate. The rotary-side joint **33** can rotate in the circumferential direction but does not move vertically relative to the attachment **31**. Hereinafter, the relative vertical movement of the fixed-side joint **32** with respect to the attachment **31** is simply referred to as vertical movement.

[0056] Since the rotary-side joint **33** is tubular, it has a coaxial through hole in the axial direction. This through hole is the first hole **330**. The rotary-side joint **33** is connected coaxially and liquid-tight to the upper end of the rotary shaft **22** of the motor **21**. By connecting the rotary shaft **22** and the rotary-side joint **33**, the fourth hole **220** of the rotary shaft **22** and the first hole **330** of the rotary-side joint **33** communicate with each other, and the rotary-side joint **33** rotates in the circumferential direction along with the rotation of the rotary shaft **22**.

[0057] For example, the rotary-side joint **33** is inserted liquid-tight into the rotary shaft **22** from above through an opening at the upper end side of the rotary shaft **22** so that the upper end of the rotary-side joint **33** protrudes upward from the rotary shaft **22**. As shown in FIGS. **3** and **4**, the rotary-side joint **33** has a flange **331** on its outer peripheral surface. The flange **331** contacts the upper end face of the rotary shaft **22** from above. By tightening the rotary-side joint **33** to the rotary shaft **22** with a screw, the rotary-side joint **33** can rotate along with the rotary shaft **22**.

[0058] As shown in FIGS. **3** and **4**, the rotary-side joint **33** has a groove **332** on its outer peripheral surface. The groove **332** runs continuously in the circumferential direction of the rotary-side joint **33**. An O-ring **262** is housed in the groove **332**. As shown in FIG. **3**, the O-ring **262** is interposed over the entire circumference between the outer peripheral surface of the rotary-side joint **33** and the inner peripheral surface of the rotary shaft **22**. Therefore, liquid leakage between the rotary-side joint **33** and the rotary shaft **22** can be prevented.

[0059] Since the fixed-side joint **32** is tubular, it has a coaxial through hole in the axial direction. This through hole is the second hole **320**. As shown in FIGS. **2-4**, at least the upper end of the fixed-side joint **32** is inserted coaxially into the third hole **310** of the attachment **31** from below so that the fixed-side joint **32** becomes coaxial with the attachment **31**. The lower surface of the fixed-side joint **32** faces the upper surface of the rotary-side joint **33**.

[0060] As shown in FIG. **3**, a DLC film **35** is provided on the outer peripheral surface of the upper end of the fixed-side joint **32**. The range where the DLC film **35** is provided will be described later.

The DLC film **35** is a diamond-like carbon film with a thickness of several microns, which is very thin compared to the thickness of the peripheral wall of the fixed-side joint **32**. However, in FIG. **3**, the DLC film **35** is shown thicker for clarity. To obtain low friction characteristics with the elastic member **36**, the hydrogen content of the DLC film **35** is preferably 40-70 atomic percent.

[0061] As shown in FIGS. **3** and **4**, the third hole **310** of the attachment **31** has a groove **34** on the inner peripheral surface. The groove **34** runs continuously in the circumferential direction of the attachment **31**. An elastic member **36** is housed in the groove **34**.

[0062] The elastic member **36** is an O-ring made of fluororubber (preferably vinylidene fluoride-based rubber (FKM)). The elastic member **36** has a circular cross-section. The elastic member **36** is interposed liquid-tight over the entire circumference between the inner peripheral surface of the third hole **310** of the attachment **31** and the DLC film **35** so that the DLC film **35** can slide vertically relative to the elastic member **36**.

[0063] The coolant that has flowed into the internal space of the fixed member **23** through the supply port **231** shown in FIG. **2** flows into the third hole **310** of the attachment **31** from the internal space of the fixed member **23**. The fixed-side joint **32** moves downward due to the hydraulic pressure of the coolant that has flowed into the third hole **310** of the attachment **31**. When the fixed-side joint **32**, which is separated from the rotary-side joint **33**, moves downward, the lower surface of the fixed-side joint **32** approaches the upper surface of the rotary-side joint **33**. By the lower surface of the fixed-side joint **32** coming into contact with the upper surface of the rotary-side joint **33** from above, the fixed-side joint **32** that has moved downward can be prevented from falling out of the attachment **31**.

[0064] For example, the hydraulic pressure of the coolant supplied from the coolant supply device is 7 MPa, and the coolant presses the lower surface of the fixed-side joint **32** against the upper surface of the rotary-side joint **33**. At this time, the lower surface of the fixed-side joint **32** and the upper surface of the rotary-side joint **33** are in close contact with each other, thus preventing liquid leakage between the lower surface of the fixed-side joint **32** and the upper surface of the rotary-side joint **33**.

[0065] When the lower surface of the fixed-side joint **32** is in contact with the upper surface of the rotary-side joint **33**, the second hole **320** of the fixed-side joint **32** communicates with the first hole **330** of the rotary-side joint **33**. The coolant that has flowed into the third hole **310** of the attachment **31** flows in this order through the third hole **310** of the attachment **31**, the second hole **320** of the fixed-side joint **32**, the first hole **330** of the rotary-side joint **33**, the fourth hole **220** of the rotary shaft **22** of the motor **21**, and the through-hole **240** of the spindle **24**. The coolant that has flowed through the through-hole **240** of the spindle **24** flows into the flow path **201** of the tool **20**. The coolant that has flowed into the flow path **201** of the tool **20** is ejected outward through the nozzle **202**. When machining the workpiece **W**, the coolant reduces the friction between the tool **20** and the workpiece **W** and washes away the chips generated during the machining of the workpiece **W**.

[0066] In a case where the supply of coolant stops while the motor **21** is rotating, the wind pressure generated by the rotation of the rotary-side joint **33** pushes the fixed-side joint **32** upwards, thus moving the fixed-side joint **32** upwards. Note that the rotary joint device **3** may include a biasing member that biases the fixed-side joint **32** upwards. The force with which the biasing member biases the fixed-side joint **32** upwards is smaller than the force with which the coolant pushes the fixed-side joint **32** downwards.

[0067] In a case where the fixed-side joint **32**, which is in contact with the rotary-side joint **33**, moves upwards, the lower surface of the fixed-side joint **32** separates from the upper surface of the rotary-side joint **33**, as shown in FIG. **3**. The separation distance between the lower surface of the fixed-side joint **32** and the upper surface of the rotary-side joint **33** is, for example, several millimeters, but it is exaggerated in FIG. **3** for clarity.

[0068] As shown in FIG. **4**, the fixed-side joint **32** has a guide receiving section **322**. The guide receiving section **322** is, for example, a notch provided at the outer edge of a flange portion of the

fixed-side joint **32**. The guide receiving section **322** is penetrated by a cylindrical guide **313**. The guide **313** extends vertically and is screwed to the attachment **31** so as to hang down from the lower surface of the flange **311** of the attachment **31**.

[0069] By the inner surface of the guide receiving section **322** contacting the outer peripheral surface of the guide **313**, the guide **313** also functions as a stopper to prevent the fixed-side joint **32** from rotating in the circumferential direction. Therefore, even if the lower surface of the fixed-side joint **32** is in contact with the upper surface of the rotary-side joint **33**, the rotation of the fixed-side joint **32** can be prevented from rotating along with the rotation of the rotary-side joint **33**.

[0070] While the rotary-side joint **33** is rotating, the upper surface of the rotary-side joint **33** slides circumferentially against the lower surface of the fixed-side joint **32**.

[0071] To reduce friction and wear caused by sliding, sliding auxiliary members **321** and **333** may be provided on the peripheral edges of the openings on the lower surface of the fixed-side joint **32** and the upper surface of the rotary-side joint **33**, respectively.

[0072] The sliding auxiliary members **321** and **333** are respectively annular and flat in the axial direction and are coaxial with the fixed-side joint **32** and the rotary-side joint **33**. Contact between the lower surface of the fixed-side joint **32** and the upper surface of the rotary-side joint **33** means contact between the sliding auxiliary member **321** and the sliding auxiliary member **333**. The sliding auxiliary members **321** and **333** are wear-resistant and made of, for example, silicon carbide.

[0073] Due to the reciprocating vertical movement of the fixed-side joint **32**, the elastic member **36** slides vertically relative to the DLC film **35**. Hereinafter, the relative vertical sliding of the elastic member **36** against the DLC film **35** will simply be referred to as the sliding of the elastic member **36**. Now, the range where the DLC film **35** is provided will be described.

[0074] As shown in FIG. **3**, the DLC film **35** is provided in the contact area **3a** and the non-contact area **3b**.

[0075] The contact area **3a** is a range on the outer peripheral surface of the fixed-side joint **32** where the elastic member **36** contacts. Due to the low friction of the DLC film **35** against the elastic member **36**, the sliding of the elastic member **36** can be smooth. Therefore, the movement of the fixed-side joint **32** in the vertical direction relative to the rotary-side joint **33** can be smooth. The contact area **3a** is located in the middle of the axial direction of the fixed-side joint **32**.

[0076] The non-contact area **3b** is a range on the outer peripheral surface of the fixed-side joint **32** where the elastic member **36** does not contact. The non-contact area **3b** is continuous with the contact area **3a** and extends from the upper end of the contact area **3a** to the upper end of the fixed-side joint **32**. Note that the non-contact area **3b** only needs to be located above the contact area **3a**, and it does not need to extend to the upper end of the fixed-side joint **32**. As shown in FIG. **4**, when the upper edge of the fixed-side joint **32** is chamfered, the DLC film **35** may or may not be provided on the chamfered portion of the fixed-side joint **32**.

[0077] There is a gap between the inner peripheral surface of the third hole **310** of the attachment **31** and the DLC film **35** for the smooth vertical movement of the fixed-side joint **32**. This gap is exaggerated in FIG. **3** for clarity. A part of the coolant that has flowed into the third hole **310** of the attachment **31** infiltrates the gap between the inner peripheral surface of the non-contact area **3b** of the third hole **310** of the attachment **31** and the DLC film **35**. Although the gap between the inner peripheral surface of the third hole **310** of the attachment **31** and the DLC film **35** is narrow, if minute foreign matter is mixed in the coolant, the foreign matter may infiltrate the gap between the inner peripheral surface of the third hole **310** of the attachment **31** and the DLC film **35**. However, due to the low friction of the DLC film **35** against the foreign matter, there the foreign matter will not hinder the movement of the fixed-side joint **32** in the vertical direction.

[0078] Since the DLC film **35** can be provided on the outer peripheral surface of the fixed-side joint **32** using a well-known DLC coating technique, there is no need to form supply holes in the attachment **31** for supplying grease between the outer peripheral surface of the fixed-side joint **32**

and the elastic member **36**. Therefore, the structure of the rotary joint device **3** is simplified.

[0079] As a result, smooth reciprocating movement of the fixed-side joint **32** and simplification of the structure can be achieved simultaneously.

[0080] Moreover, since the DLC film **35** is highly resistant to coolant and wear, it is maintenance-free.

[0081] Since the frictional force generated between the DLC film **35** and the elastic member **36** is small, the rotary joint device **3** can be designed so that the force with which the elastic member **36** contacts the DLC film **35** is large. Since the elastic member **36** is an O-ring, the compression rate of the elastic member **36** can be increased so that the force with which the elastic member **36** contacts the DLC film **35** is large. Therefore, liquid leakage between the inner peripheral surface of the third hole **310** of the attachment **31** and the outer peripheral surface of the fixed-side joint **32** can be reliably prevented.

[0082] The compression rate of the elastic member **36** is preferably 10-30%. If the cross-sectional diameter of the elastic member **36** is 1.9 mm, the inner diameter of the third hole **310** of the attachment **31** is 12 mm, and the outer diameter of the fixed-side joint **32** is 9 mm, the compression rate is about 21% ($=\{1.9-(12-9)/2\}+1.9\times100$).

[0083] The DLC film **35** may not necessarily be provided in the range on the outer peripheral surface of the fixed-side joint **32** from the lower end of the contact area **3a** to the lower end of the fixed-side joint **32**. This is because the coolant will not infiltrate (and hence foreign matter mixed in the coolant will not infiltrate) the gap between the inner peripheral surface of the third hole **310** of the attachment **31** and the DLC film **35** if the DLC film is provided in this range. At least the surface roughness (arithmetic average roughness Ra) of the fixed-side joint **32**'s outer peripheral surface in the contact area **3a** and the non-contact area **3b** is desirably 0.2 or less. If the arithmetic average roughness exceeds 0.2, the low friction of the DLC film **35** may be insufficient due to the surface roughness of the contact area **3a** and the non-contact area **3b**.

[0084] Herein, it is considered to provide a diamond film on the outer peripheral surface of the fixed-side joint **32** instead of the DLC film **35**. The diamond film is more wear-resistant than the DLC film **35**.

[0085] However, generally, the temperature during the deposition of a CVD diamond film using a hot filament method, which can also form films on the external surface of the joint, is higher than the temperature during the DLC film **35** deposition. A fixed-side joint **32** exposed to the high temperature (about 600-700° C.) during diamond film deposition is prone to distortion. On the other hand, the fixed-side joint **32** is less prone to distortion even if exposed to the low temperature (200° C. or lower) during DLC film **35** deposition. Furthermore, thickness control during diamond film deposition is difficult, but thickness control during DLC film **35** deposition is easy. Therefore, adopting the DLC film **35** contributes to the improvement of the rotary joint device **3**'s accuracy. Additionally, the diamond film is more expensive than the DLC film **35**. Therefore, adopting the DLC film **35** contributes to the reduction of manufacturing costs for the rotary joint device **3**.

[0086] Moreover, if the foreign matter mixed in the coolant will not infiltrate the gap between the inner peripheral surface of the third hole **310** of the attachment **31** and the outer peripheral surface of the fixed-side joint **32** and not hinders the vertical movement of the fixed-side joint **32**, it is unnecessary to provide the DLC film **35** in the non-contact area **3b**.

[0087] The elastic member **36** is not limited to an O-ring made of fluororubber. The elastic member **36** may be made of any material that has a low coefficient of friction with the DLC film **35**, is not easily eroded by the coolant, and can function as a packing.

[0088] The rotary joint device **3** is not limited to the configuration provided in the machine tool **1**. The attachment **31** is not limited to being fixed to the fixed member **23**. The rotary-side joint **33** is not limited to being connected to the rotary shaft **22** of the motor **21**. The direction in which the fixed-side joint **32** and the rotary-side joint **33** face each other is not limited to the vertical direction. The reciprocating movement direction of the fixed-side joint **32** is not limited to the

vertical direction.

[0089] While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents.

Claims

1. A rotary joint device comprising: a rotary-side joint having a tubular shape and a first hole through which liquid is to flow in an axial direction, the rotary-side joint being rotatable in a circumferential direction; a fixed-side joint having a tubular shape and an end face at one end facing an end face at one end of the rotary-side joint, the fixed-side joint being configured to reciprocate in the axial direction of approaching and separating from the rotary-side joint, the fixed-side joint further having a second hole in the axial direction that communicates with the first hole of the rotary-side joint in a state where the fixed-side joint is in contact with the rotary-side joint; an attachment having an inner peripheral surface, the attachment being fixable to another member, the inner peripheral surface defining a third hole through which the liquid is to flow and into which the other end of the fixed-side joint is inserted; a DLC (diamond-like carbon) film provided on an outer peripheral surface of the fixed-side joint; and an elastic member interposed liquid-tight between the inner peripheral surface of the attachment and the DLC film, the elastic member being slidable with respect to the DLC film when the fixed-side joint moves in the axial direction with respect to the rotary-side joint, wherein the DLC film is provided in a contact area where the elastic member contacts on the outer peripheral surface.
2. The rotary joint device according to claim 1, wherein the DLC film is provided in the contact area and a non-contact area on the outer peripheral surface where the elastic member does not contact, and wherein the non-contact area is located on the end face side of the other end of the fixed-side joint relative to the contact area.
3. The rotary joint device according to claim 1, wherein the elastic member is an O-ring made of fluororubber.
4. A motor comprising: a rotary shaft rotatable in the circumferential direction and having a tubular shape and a through hole through which liquid flows in an axial direction, wherein the rotary shaft is connected with the rotary-side joint of the rotary joint device according to claim 1 at another end of the rotary-side joint which is opposite to the one end of the rotary-side joint to integrally rotate with the rotary-side joint while the through hole and the first hole of the rotary-side joint communicate with each other.
5. A machine tool for processing a workpiece while discharging coolant from a nozzle using a tool provided with the nozzle, the machine tool comprising: the rotary joint device according to claim 1; a fixed member to which the attachment of the rotary joint device is fixed; and a motor including a rotary shaft rotatable in the circumferential direction and having a tubular shape and a fourth hole through which liquid is to flow in an axial direction, wherein the rotary shaft is connected with the rotary-side joint of the rotary joint device at another end of the rotary-side joint which is opposite to the one end of the rotary-side joint to integrally rotate with the rotary-side joint while the through hole and the first hole of the rotary-side joint communicate with each other, wherein rotation of the rotary shaft is to be transmitted to the tool to rotate the tool, wherein the liquid is to flow through the third hole, the second hole, the first hole, and the fourth hole in this sequence to

be ejected through the nozzle of the tool, wherein the DLC film being provided in the contact area where the elastic member contacts on the outer peripheral surface.
